NRES 746, Week 13, Lecture 1: Partial Pooling by Hand

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

Instructions: Work alone (about 5 minutes), then compare with a neighbor. Keep laptops closed.

You swab frogs for the chytrid fungus Bd in many ponds and record each frog's infection load ($\log_{10}$ zoospore equivalents). A random intercept model fit to all the ponds gives an overall mean load $\hat{\mu} = 3.0$, an among-pond SD $\sigma_a = 0.5$, and a within-pond (frog-to-frog) SD $\sigma_e = 1.0$. The model's estimate for pond j is

$$\hat{a}_j = \hat{\mu} + w_j(\bar{y}_j - \hat{\mu}), \qquad w_j = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_e^2/n_j}$$

where $\bar{y}_j$ is the mean load of the n_j frogs swabbed in that pond.

Pond	Frogs swabbed (n_j)	Mean load ($\bar{y}_j$)
A	4	4.2
B	16	4.0

1a. $w_A =$ 0.5 $\quad w_B =$ 0.8

$$w_A = \frac{0.25}{0.25 + 1/4} = \frac{0.25}{0.5} = 0.5, \qquad w_B = \frac{0.25}{0.25 + 1/16} = \frac{0.25}{0.3125} = 0.8$$

1b. $\hat{a}_A =$ 3.6 $\quad \hat{a}_B =$ 3.8

$$\hat{a}_A = 3.0 + 0.5(4.2 - 3.0) = 3.6, \qquad \hat{a}_B = 3.0 + 0.8(4.0 - 3.0) = 3.8$$

1c. Which pond would you report as more heavily infected? Explain why the model's ranking differs from the ranking of the raw means.

Pond B (3.8 vs. 3.6). With only 4 frogs, pond A's mean has sampling variance $\sigma_e^2/n = 0.25$, as large as the among-pond variance, so half of its distance from the overall mean is treated as noise. Pond B's 16 frogs make its mean precise enough to keep 80% of its distance.